

Rafael Augusto Macchia

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Spanish (Native), English (Fluent)



SUMMARY

Astrophysics, Robotics & CS scholar. Dual-enrollment A.A. graduate experienced in Java/Python development, CAD, physics research, and robotics leadership.

EDUCATION

- **Miami Dade College (DE):** A.A. Degree | Coursework: Python, Chem, Physics, Calculus, AICE
- **iPrep Academy:** High School Diploma | GPA: ~3.5

EXPERIENCE

- **Robotics Intern** | CIEE / FingerVision (Tokyo) (Summer 2026): Worked with visual-tactile robotic sensors & control algorithms.
- **Cybersecurity & Software** | Cybersecurity & Systems Developer | Personal Projects (2025–2026): Programmed network diagnostic tools, security scripts, and custom OOP applications in Python and Java.
- **Captain & Lead Developer** | Venom Byte FTC Robotics (2022–2025): Directed goBILDA CAD builds, Java software, & strategy; won regional awards.

PROJECTS & ATHLETICS

- **Taekwondo State Champion** (2015–Pres.): 11+ years training, Black Belt (5th Dan), State Champion.
- **FTC Robotics Lead Builder & Programmer (2022–Pres.):** Custom CAD mechanical design, Java autonomous navigation algorithms & competitive strategy.
- **3D Gravitational N-Body Simulator (2025–Pres.):** Computational physics engine, 3D vector calculus & numerical differential equation solvers in Python.
- **CIEE Tokyo Robotics Internship (2026):** Visual-tactile sensor integration & camera-based tactile feedback systems with FingerVision in Japan.
- **Interactive Portfolio Website (2025–Pres.):** Custom glassmorphic UI design, JavaScript scroll triggers & automated GitHub Actions deployment.

SKILLS & TOOLS

- **Languages & Tech:** Java, Python, JavaScript, HTML/CSS, OOP, goBILDA, Tactile Sensors, Git/GitHub, VS Code, Linux CLI